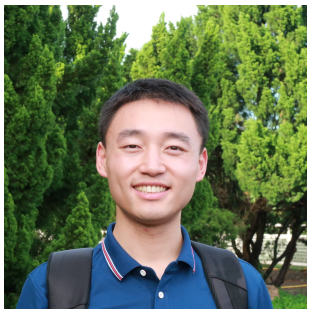

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## 研究方向

**优化理论与算法及其在科学与工程计算中的应用**. 目前研究聚焦计算量子化学与计算材料科学，研究主题包括电子激发态计算、电子能量景观分析、强关联电子体系基态计算、晶体结构弛豫等.

## 教育经历

| | | | |
|---|---|---|---|
| 2019.09 - 2024.06 | 中国科学院大学 | 理学博士 (计算数学专业) | 指导教师: 刘　歆 研究员 |
| 2015.09 - 2019.06 | 同济大学 | 理学学士 (数学与应用数学专业) | 指导教师: 殷俊锋 教授 |

## 工作经历

| | | | |
|---|---|---|---|
| 2024.09 - 2026.08 | 巴黎理工学院, 法国国立路桥学校 | 博士后研究员 | 合作导师: Eric Cancès 教授 |
| 2024.07 - 2024.08 | 香港理工大学, 应用数学系 | 研究助理 | 合作导师: 张在坤 助理教授 |

## 已发表论文与预印本

*: equal contribution, †: corresponding author

1. *L. Grazioli, Y. Hu, T. Nottoli, F. Lipparini, and E. Cancès. Critical point search and linear response theory for computing electronic excitation energies of molecular systems. Part II: CASSCF. arXiv preprint arXiv:2604.13753, 2026. https://arxiv.org/abs/2604.13753.
2. Y. Wang, Y. Hu, and X. Liu. Lower bound on the representation complexity of antisymmetric tensor product functions. *Science China Mathematics*, 2026. DOI: 10.1007/s11425-025-2547-3.
3. *L. Grazioli, Y. Hu, and E. Cancès. Critical point search and linear response theory for computing electronic excitation energies of molecular systems. Part I: General framework, application to Hartree-Fock and DFT. *The Journal of Chemical Physics*, 2026, 164(6): 064101 (22 pages). DOI: 10.1063/5.0302841.
4. †Y. Hu and L. Grazioli. Constrained dynamics for searching saddle points on general Riemannian manifolds. arXiv preprint arXiv:2601.03931, 2026. https://arxiv.org/abs/2601.03931.
5. Y. Hu, M. Li, X. Liu, and C. Meng. Sampling-based approaches for multi-block optimization problems over transport polytopes. *Mathematics of Computation*, 2025, 94(353): 1281-1322. DOI: 10.1090/mcom/3989.
6. Y. Hu and X. Liu. The exactness of the $\ell_1$ penalty function for a class of mathematical programs with generalized complementarity constraints. *Fundamental Research*, 2024, 4(6): 1459-1464. DOI: 10.1016/j.fmre.2023.04.006.
7. Y. Hu, J. Yin, X. Gao, X. Liu, and H. Song. Projected gradient descent algorithm for *ab initio* crystal structure relaxation under a fixed unit cell volume. *Physical Review B*, 2024, 109(22): 224109 (14 pages). DOI: 10.1103/PhysRevB.109.224108.
8. Y. Hu and X. Liu. The convergence properties of infeasible inexact proximal alternating linearized minimization. *Science China Mathematics*, 2023, 66(10): 2385-2410. DOI: 10.1007/s11425-022-2074-7.
9. Y. Hu, H. Chen, and X. Liu. A global optimization approach for multimarginal optimal transport problems with Coulomb cost. *SIAM Journal on Scientific Computing*, 2023, 45(3): A1214-A1238. DOI: 10.1137/21M1455164.
10. Y. Hu, X. Gao, Y. Zhao, X. Liu, and H. Song. Force-based gradient descent method for *ab initio* atomic structure relaxation. *Physical Review B*, 2022, 106(10): 104101 (10 pages). DOI: 10.1103/PhysRevB.106.104101.

## 知识产权与软件

1. 高兴誉, 刘歆, 胡雨宽, 宋海峰, 陈欣, 王越超, 王丽芳. 基于非单调梯度型算法的晶体结构弛豫软件 ProME-SuRe: 2023SR1558824. 软件著作权. 2023-12-04.
2. 高兴誉, 刘歆, 宋海峰, 胡雨宽, 陈欣, 王越超, 方俊, 王丽芳, 张乐. 固定晶格体积晶体结构弛豫的计算方法及装置: 202211210741.3. 发明专利. 2023-07-21.
3. 高兴誉, 刘歆, 宋海峰, 胡雨宽, 方俊, 杨真, 赵亚帆, 王丽芳, 刘海风. 原子结构弛豫的非单调线搜索方法及装置: 202111534901.5. 发明专利. 2022-12-02.

## 部分参研项目

| | | |
|---|---|---|
| 2024.09 - 2026.08 | ERC, EU Horizon 2020 | "Extreme-scale Mathematically-based Computational Chemistry" |
| 2022.12 - 2023.11 | JKW "16X 工程" | "原子结构弛豫优化算法研究与程序研制" |
| 2019.09 - 2021.12 | 国防基础科研核科学挑战专题 | "晶体结构弛豫的高效局域优化算法" |

## 部分学术报告

| | | |
|---|---|---|
| 2026.06 | The 16th International Conference on Scientific Computation and Differential Equations (SciCADE 2026) | *Edinburgh, UK* |
| 2026.03 | The 96th Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM 2026) | *Stuttgart, Germany* |
| 2025.08 | The 5th ABACUS Developers Conference (Poster) | *Beijing, China* |
| 2025.08 | The 15th International Conference on Optimization and Numerical Linear Algebra | *Harbin, China* |
| 2025.07 | The 8th International Conference on Continuous Optimization (ICCOPT 2025) | *Los Angeles, USA* |
| 2025.06 | The 22nd Conference on Advances in Continuous Optimization (EUROPT 2025) | *Southampton, UK* |
| 2024.07 | The 1st Forum for Ph.D. Students in Computational Mathematics | *Beijing, China* |
| 2023.10 | The 21st Annual Meeting of China Society for Industrial and Applied Mathematics | *Kunming, China* |
| 2023.08 | The 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023) | *Tokyo, Japan* |
| 2023.08 | The 14th International Conference on Optimization and Numerical Linear Algebra | *Taiyuan, China* |

## 部分获奖经历

| | | |
|---|---|---|
| 2025 | **中国科学院优秀博士学位论文** | 中国科学院 |
| 2024 | **中国科学院院长奖学金特别奖** | 中国科学院 |
| 2023 | Financial Support Program | ICIAM 2023 |
| 2023 | 中国科学院朱李月华奖学金 | 中国科学院大学 |
| 2023 | 中国科学院数学与系统科学研究院院长奖学金特等奖 | 中国科学院数学院 |
| 2023 | 北京数学会首届青年优秀论文 | 北京数学会 |
| 2022 | 中国科学院数学与系统科学研究院华罗庚奖学金 | 中国科学院数学院 |
| 2021 | 中国科学院数学与系统科学研究院华罗庚奖学金 | 中国科学院数学院 |

## 部分学术服务

- 会议组织

| | | | |
|---|---|---|---|
| 2025.06 | EUROPT 2025 | Session Chair for "Advances in Manifold Optimization" | *Southampton, UK* |

- 期刊审稿

IEEE Transactions on Signal Processing
Journal of Computational and Applied Mathematics
The Innovation
Numerical Functional Analysis and Optimization
npj Unconventional Computing